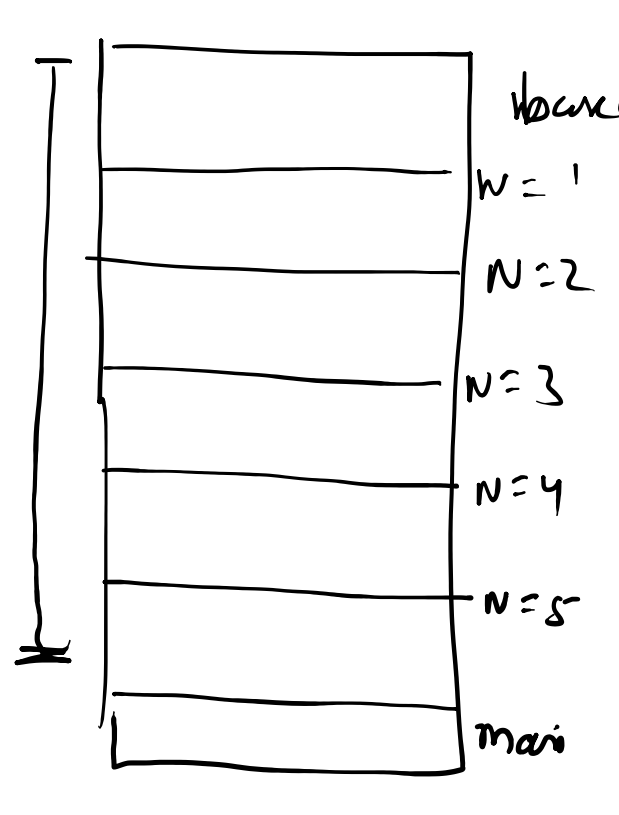
Lec-15

03/07/26 11:17 AM

```
public static int fac(int n) {
    // Base case
    if (n == 0) {
        return 1;
    }
    int fn = fac(n - 1);// SP
    return fn * n;
}
```

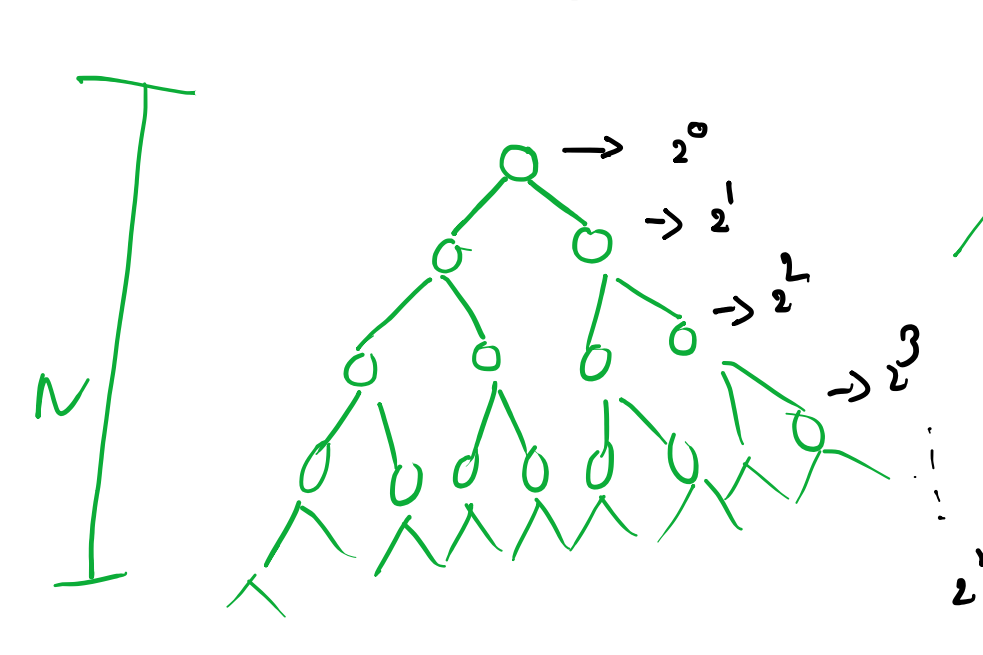
} - for(i=0; i<n; i++){

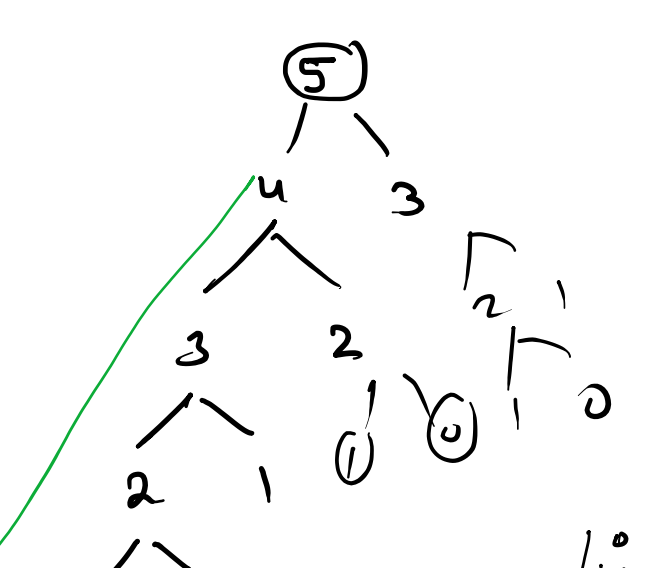
$n+n+n+\ldots+n$ upto n time } n^2


Heap
N=1
N=2
N=3
N=4
N=5
main

```
public static int fib(int n) {
    if (n == 0 || n == 1) {
        return n;
    }
    int f1 = fib(n - 1);
    int f2 = fib(n - 2);
    return f1 + f2;
}
```

O(1)


N
$\to 2^0$
$\to 2^1$
$\to 2^2$
$\to 2^3$
2^n


5
4
3
3
2
2
1
2
1
1
0
1
0
1
0

1, 10, 100, 1000 ⑩$^3 = 10^3 = 1000$

$a + ar + ar^2 + ar^3 + \ldots + ar^n$

$a \times [r^N - 1] / (r - 1)$

$2^0 + 2^1 + 2^2 + 2^3 + \ldots + 2^n = 2^0 \frac{[2^{N+1} - 1]}{2-1} = 2^{N+1} - 1 = 2 \times 2^n - 1$

T.C $= 2^n$

```
public static int fac(int n) {
    // Base case
    if (n == 0) {
        return 1;
    }
    int fn = fac(n - 1);// SP
    return fn * n;
}
```

↳ pow

O(1)

①

$T(N) = T(N-1) + 1$

$T(N-1) = T(N-2) + 1$

$T(N-2) = T(N-3) + 1$

$T(N-3) = T(N-4) + 1$

$T(N-4) = T(N-5) + 1$

$T(1) = 1$

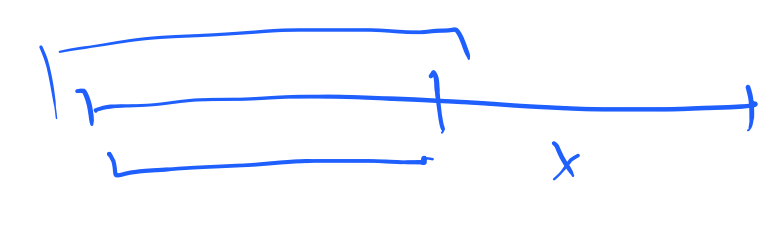
$T(n) = N$

$T(N) = N$

$O(N)$

① Merge ② Quick ③ fibo ④ Binary search

	Merge	Quick
worst case	$N\log(N)$	n^2
Avg	$N\log(N)$	$N\log(N)$
best	$N\log(N)$	$N\log(N)$
	$O(n)$	$O(1)$



$T(N) = T(N/2) + 1$

$T(N/2) = T(N/2^2) + 1$

$T(N/2^2) = T(N/2^3) + 1$

$T(N/2^3) = T(N/2^4) + 1$

$T(N/2^k = 1) = 1$

$T(n) = K$

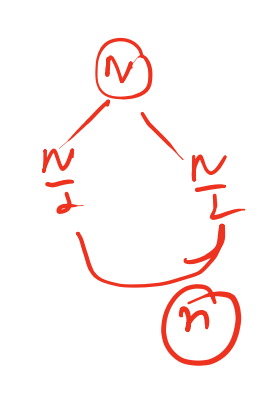
$\boxed{= \log_2 N}$

$\frac{N}{2^k} = 1$

$2^k = N$

$K = \log_2 N$

$T(n) = T(\frac{N}{2}) + T(N/2) + n$



$T(n) = 2T(N/2) + N$

$2\,T(N/2) = 2^2 T(N/2^2) + \frac{N}{2} \times 2$

$2^2\,T(N/2^2) = 2^3 T(N/2^3) + \frac{N}{2^2} \times 2^2$

$2^3\,T(N/2^3) = 2^4 T(N/2^4) + \frac{N}{2^3} \times 2^3$

$2^4\,T(N/2^4) = 2^5 T(N/2^5) + N/2^4 \times 2^4$

$2^k\,T(N/2^k) = \frac{N}{2^k} \times 2^k$

$\frac{N}{2^k} = 1$

$T(n) = K \times N$

$\log(N)$

$\boxed{= N \times \log(N)}$

```
public static int fac(int n) {
    // Base case
    if (n == 0) {
        return 1;
    }
    int fn = fac(n - 1);// SP
    return fn * n;
}
```

$T(N) = T(N-1) + T(N-2) + 1$

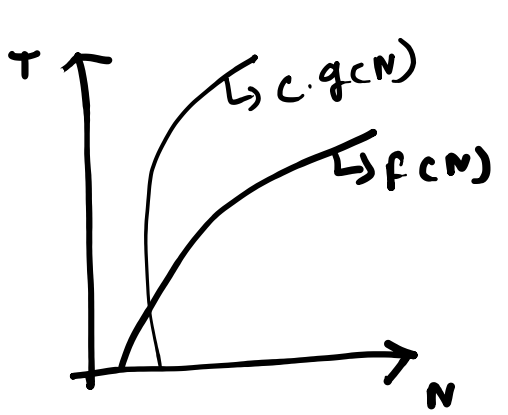
$T(N-1) = T(N-2) + T(n-3) + 1$

$T(n-2) = T(n-3) + T(n-4) + 1$

$T(n-3) = T(n-4) + T(n-5) + 1$

$T(n-4) = T(n-5) + T(n-6) + 1$

$T(1) = 1$



$f(n) \le c.g(n)$

$f(n) \to O(g(n))$

$T(n) = T(n-1) + T(n-2) + 1$

$T(n-1) + T(n-2) + 1 \le T(n-1) + T(n-1) + 1$

$T(n) = 2T(n-1) + 1$

$2\,T(n-1) = 2^2 T(n-2) + 1 \times 2$

$2^2 T(n-2) = 2^3 T(n-3) + 1 \times 2^2$

$2^3 T(n-3) = 2^4 T(n-4) + 1 \times 2^3$

$2^{N-1} T(1) = 1 \times 2^{N-1}$

$(N - (n-1))$

$T(n) = 1 + 2 + 2^2 + 2^3 + \ldots + 2^{N-1}$

$1[2^N - 1] / 2 - 1 = 2^N - 1$

$O(2^n)$

1 2 3 4 5

10, 100

5 4 3 2 1

```
public static void Sort(int[] arr, int si, int ei) {
        if (si >= ei) {
            return;
        }
        int idx = Partition(arr, si, ei);
        Sort(arr, si, idx - 1);
        Sort(arr, idx + 1, ei);
    }
    public static int Partition(int[] arr, int si, int ei) {
        int item = arr[ei];
        int idx = si;
        for (int i = si; i < ei; i++) {
            if (arr[i] < item) {
                int temp = arr[i];
                arr[i] = arr[idx];
                arr[idx] = temp;
                idx++;
            }
        }
        int temp = arr[ei];
        arr[ei] = arr[idx];
        arr[idx] = temp;
        return idx;
    }
```

[1, 2, 3, 4, 5]

1234

123

4+3+2+1

$\frac{n \times (n-1)}{2}$

n^2

[1, 2, 3, 4, 5]

0 1 2 3 4

1 2 5 4 3

$a^N \to (a^{N/2}) \times a^{N/2}$

$3^9 = 3^4 \times 3^4 \times 3$

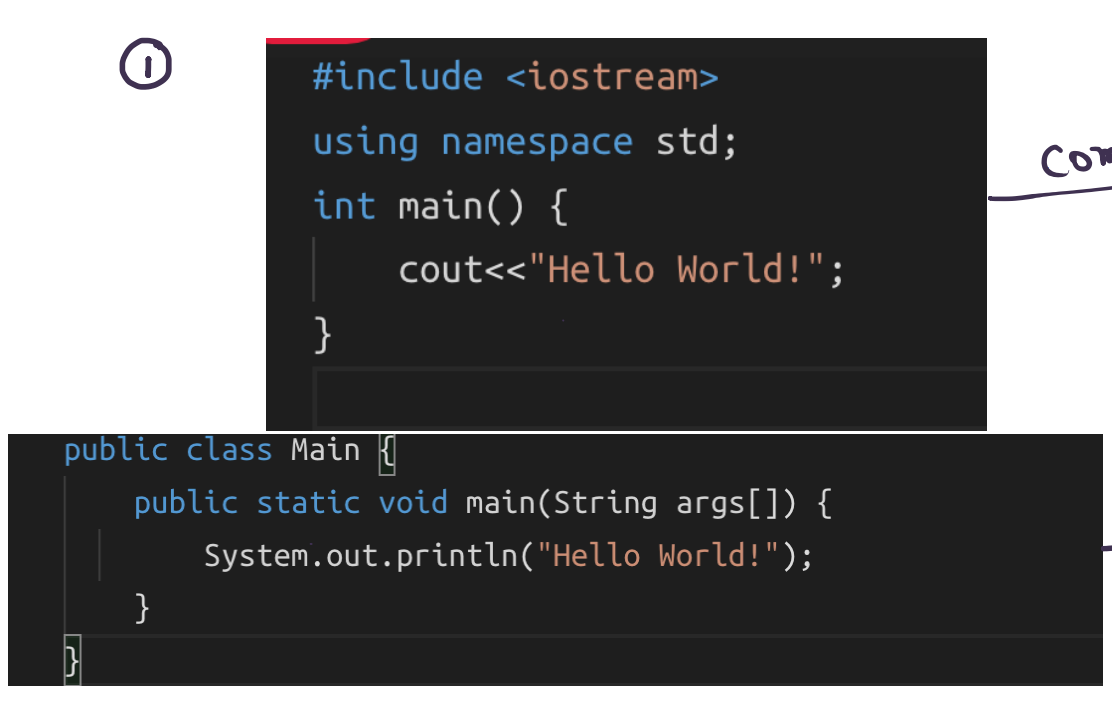
3^8

$3^{12} \to 3^{11}$

$(3^6) \times 3^6 = 3^{12}$

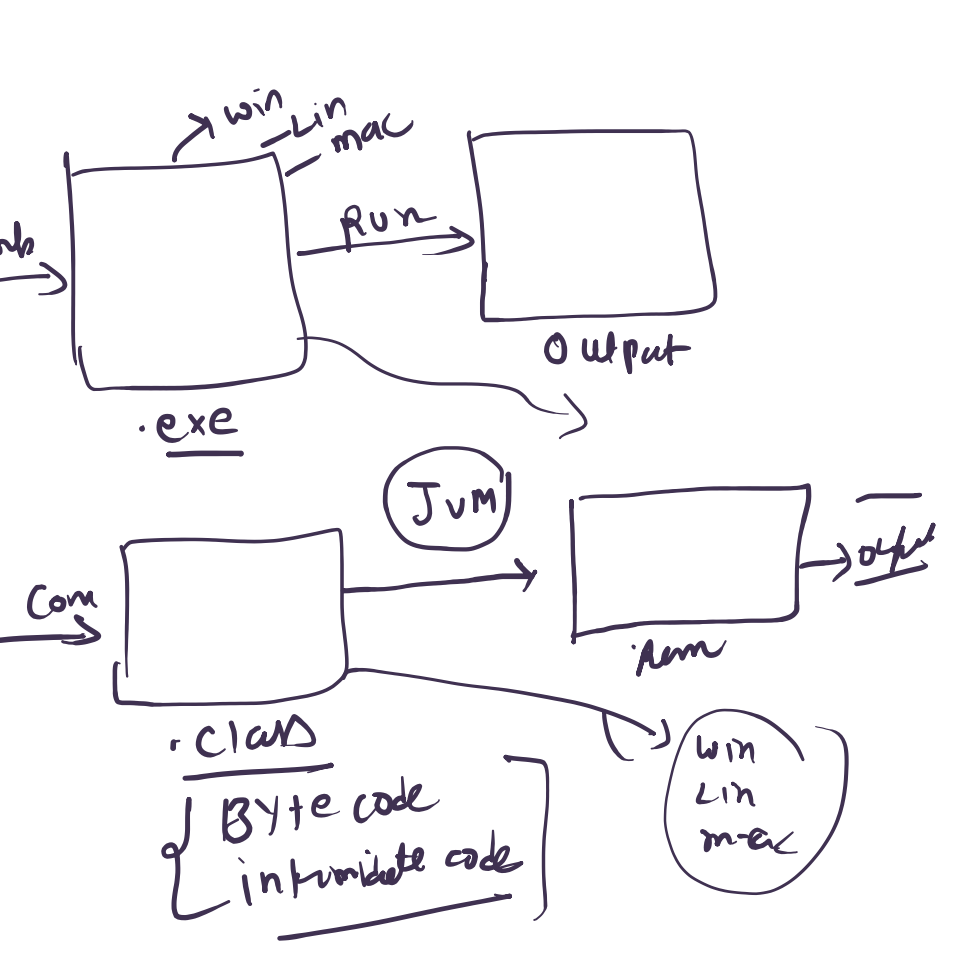
---

①

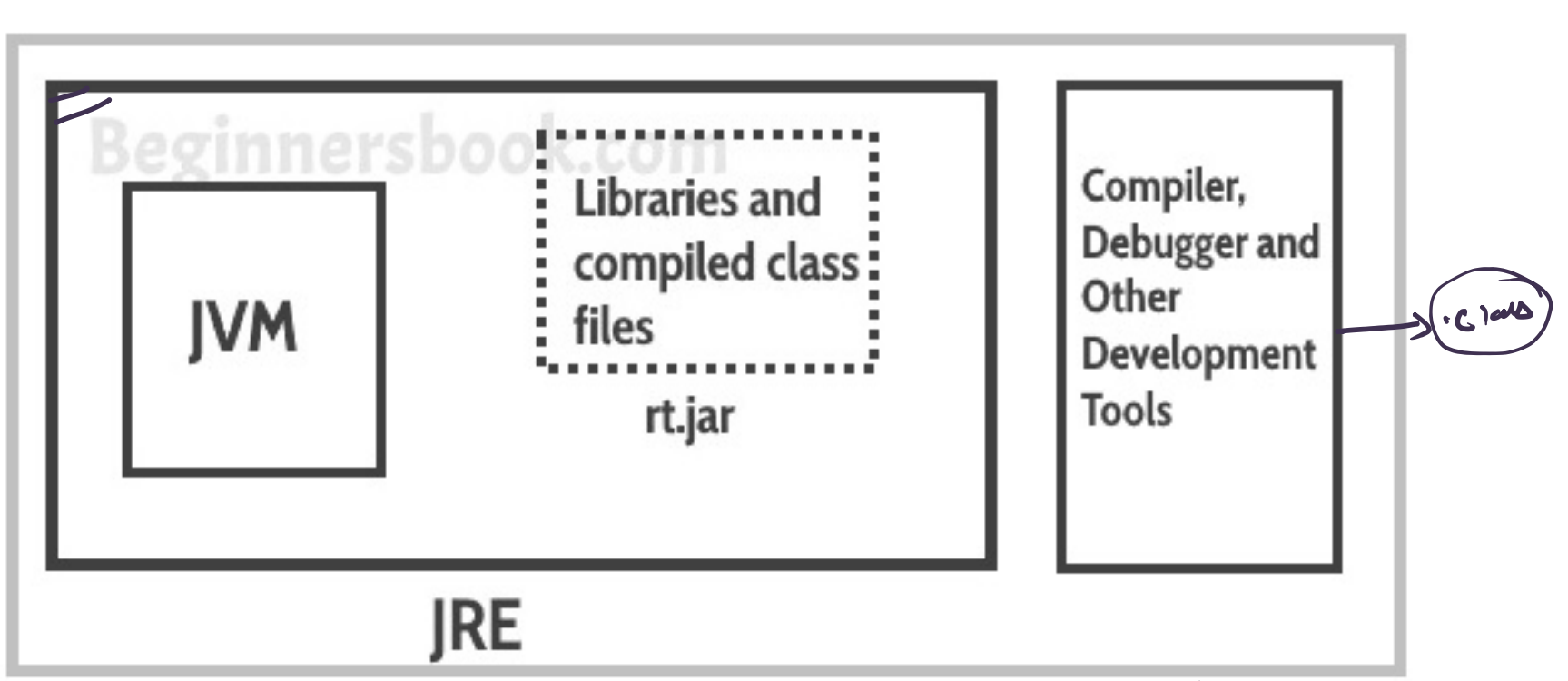


```
#include <iostream>
using namespace std;
int main() {
    cout<<"Hello World!";
}
```

```
public class Main {
    public static void main(String args[]) {
        System.out.println("Hello World!");
    }
}
```


Comp
win
Lin
mac
Run
Output
.exe
Com
JVM
output
Ram
.class
win
Lin
mac

.class { Byte code, intermidate code }


Beginnersbook.com
JVM
Libraries and compiled class files
rt.jar
Compiler, Debugger and Other Development Tools
.class
JRE
Java Development Kit(JDK)

|| JDK

I eal

VVM

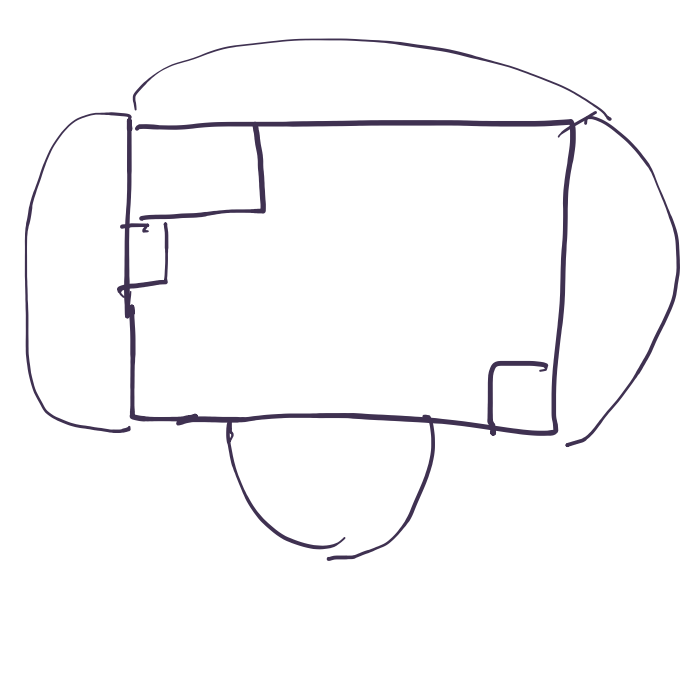
OOP

Non AH VCUB

---

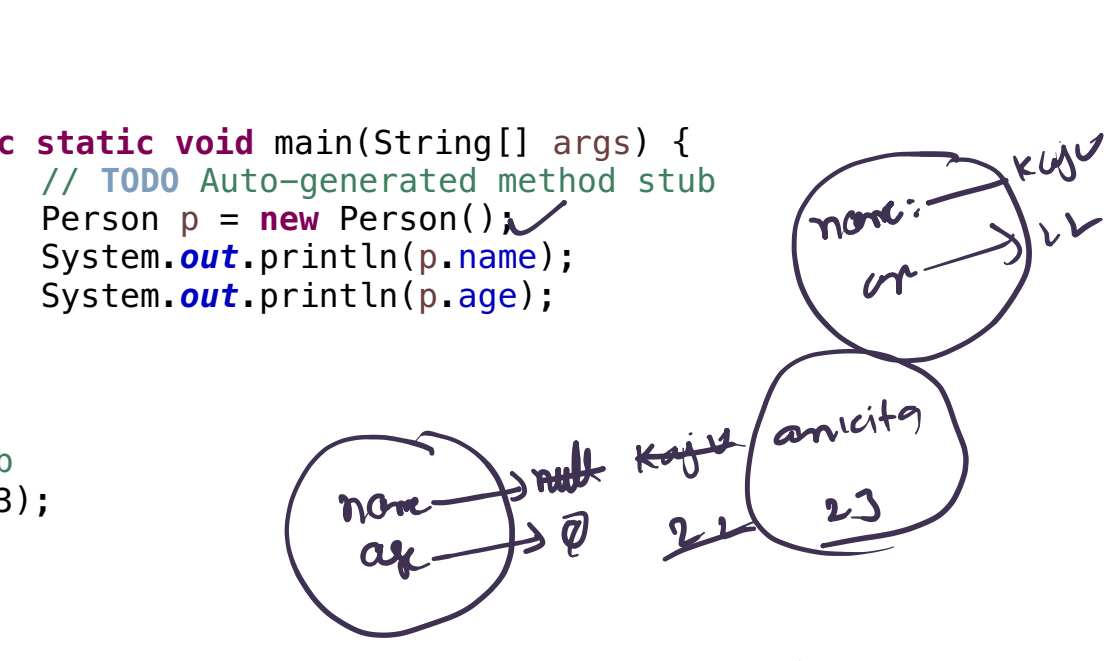
class

↳ It is blue print



```
public class Student {
    String name;
    int age;

    public void Intro_yourSelf() {
        System.out.println("My name is " + name + " and age is " + age);
    }

}
```

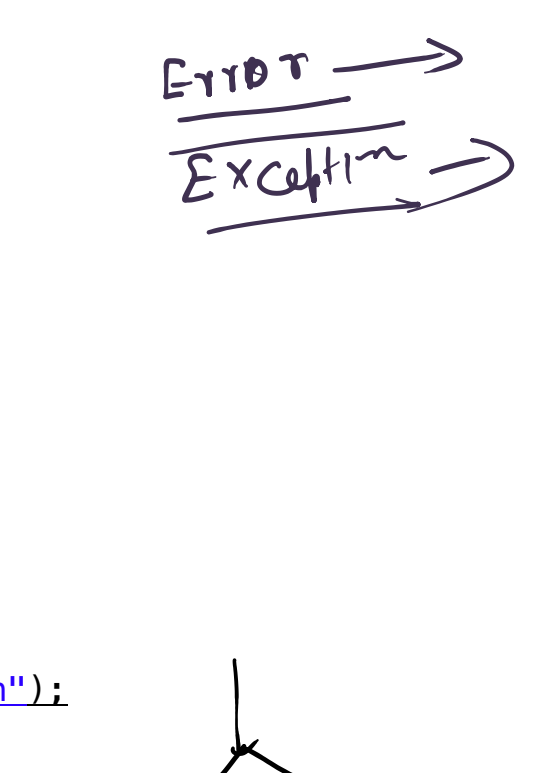
```
public class Student_Client {

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Student s = new Student();

    }

}
```


name
age
Intro
main
S=2K
name: null
age = 0

```
public class Person {
    String name = "Kaju";
    int age = 22;
}
```

```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        Person p = new Person();
        System.out.println(p.name);
        System.out.println(p.age);

    }
```

```
public static void main(String[] args) {
        // TODO Auto-generated method stub
        Person p = new Person("ankita", 23);
        System.out.println(p.name);
        System.out.println(p.age);

    }
```


name
age
Kaju
22
name
age
~~null~~
~~Kaju~~
0
22
ankita
23

public

① Private

classX

② Protected

④ default

Error →

Exception →

```
public void setAge(int age) {
        if (age < 0) {
            throw new Exception("Bklol age -ve nhi hota h");
        }
        this.age = age;
    }
```

मैं परेशान हूँ तो सबको परेशान करेंगा Throws

मैं परेशान हूँ लेकिन मैं किसी और परेशान नहीं करेंगा। } Try catch